

Real Analysis

Satoshi Fukuda

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Outline

- 1 Inner Product, Norm, and Distance
 - 2 Open and Closed Sets
 - 3 Sequences and Limits
 - 4 Continuous Functions
 - 5 Completeness of a Metric Space and Contraction Mapping Theorem
- The focus/purpose of the course is to try to understand the core **abstract** ideas/concepts in these topics.
 - Question: How will/can these abstract concepts be used throughout the first-year coursework and subsequent research?

Outline

Overview of how real analysis can/will be used

① Inner Product, Norm, and Distance

- ▶ Geometry of $\mathbb{R}^n$ and function spaces (L^2): length, angle, orthogonality.
- ▶ Projection theorem $\rightarrow$ OLS and $\mathbb{E}[Y \mid X]$ as orthogonal projections.
- ▶ Separation/supporting-hyperplane theorems $\rightarrow$ Second Welfare Theorem, no-arbitrage pricing (existence of state prices).

② Open and Closed Sets

- ▶ Foundation for limits and continuity.
- ▶ Abstracting “collections of sets”: a *topology* (open sets); a σ -*algebra* of measurable sets carrying a probability measure.

③ Sequences and Limits

- ▶ Convergence: Cauchy sequences ($\rightarrow$ completeness), Bolzano–Weierstrass (bounded $\Rightarrow$ convergent subsequence $\rightarrow$ compactness).
- ▶ Laws of Large Numbers, Central Limit Theorem, ...

Outline

Overview of how real analysis can/will be used

④ Continuous Functions

- ▶ Weierstrass Theorem: a continuous function on a compact (closed & bounded) set attains a max/min.
- ▶ Berge's Maximum Theorem: value function V continuous, policy correspondence “upper-hemi-” continuous—backbone of finding V and of comparative statics.
- ▶ Existence of competitive/Nash equilibria via fixed points (Brouwer/Kakutani: compactness + continuity + convexity).
- ▶ Choosing a consumption function $c : [0, \infty) \rightarrow \mathbb{R}$ instead of $(c_1, c_2) \in \mathbb{R}^2$, value function $V : \mathbb{R}^n \rightarrow \mathbb{R}$, a random variable $X : \Omega \rightarrow \mathbb{R}, \dots$
- ▶ Continuous Mapping Theorem (probability).

⑤ Completeness of a Metric Space and Contraction Mapping Theorem

- ▶ Cauchy $\Rightarrow$ convergent
- ▶ In $\mathbb{R}$: Existence of sup / inf
- ▶ Objective: Contraction Mapping Theorem $\rightarrow$ existence & uniqueness of value function V and convergence of value-function iteration; Blackwell's sufficient conditions (monotonicity + discounting).

Inner Product

Definition 7.1

For two points $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$, the inner product (or scalar product) $\langle \mathbf{a}, \mathbf{x} \rangle$ is the number defined as

$$\langle \mathbf{a}, \mathbf{x} \rangle := \sum_{i=1}^n a_i x_i.$$

$$\left\langle \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \right\rangle = \sum_{i=1}^n a_i x_i.$$

Proposition 7.1 (Exercise)

The inner product $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the following properties.

- ① $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$.
- ② $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
- ③ $\langle \mathbf{a}, \mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{a} \rangle$.
- ④ $\langle \mathbf{a}, \lambda \mathbf{x} \rangle = \lambda \langle \mathbf{a}, \mathbf{x} \rangle$.
- ⑤ $\langle \mathbf{a}, \mathbf{x} + \mathbf{y} \rangle = \langle \mathbf{a}, \mathbf{x} \rangle + \langle \mathbf{a}, \mathbf{y} \rangle$.

Inner Product

Proof/Solution:

① $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{i=1}^n \underbrace{x_i^2}_{\geq 0} \geq 0.$

② If $\mathbf{x} = \mathbf{0}$, then $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{i=1}^n \underbrace{x_i^2}_{=0} = 0$. Conversely, if $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{i=1}^n \underbrace{x_i^2}_{\geq 0} = 0$, then $x_i^2 = 0$,

i.e., $x_i = 0$, for all $i \in \{1, \dots, n\}$. Thus, $\mathbf{x} = \mathbf{0}$.

③

$$\langle \mathbf{a}, \mathbf{x} \rangle = \sum_{i=1}^n a_i x_i = \sum_{i=1}^n x_i a_i = \langle \mathbf{x}, \mathbf{a} \rangle.$$

④

$$\langle \mathbf{a}, \lambda \mathbf{x} \rangle = \sum_{i=1}^n a_i (\lambda x_i) = \lambda \sum_{i=1}^n a_i x_i = \lambda \langle \mathbf{a}, \mathbf{x} \rangle.$$

⑤

$$\begin{aligned} \langle \mathbf{a}, \mathbf{x} + \mathbf{y} \rangle &= \sum_{i=1}^n a_i (x_i + y_i) \\ &= \sum_{i=1}^n a_i x_i + \sum_{i=1}^n a_i y_i = \langle \mathbf{a}, \mathbf{x} \rangle + \langle \mathbf{a}, \mathbf{y} \rangle. \end{aligned}$$

Inner Product

Remark 7.2

- 1 A vector space X is an *inner product space* if it is endowed with the operation of an inner product $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{R}$ satisfying the properties in Proposition 7.1.
- 2 Statistics/Econometrics. Consider the set of square-integrable functions on $\mathbb{R}$:

$$L^2 := \left\{ \overbrace{f \in \mathbb{R}^{\mathbb{R}}}^{f: \mathbb{R} \rightarrow \mathbb{R}} \mid \int_{-\infty}^{\infty} |f(x)|^2 dx < \infty \right\}.$$

One can introduce an inner product $\langle \cdot, \cdot \rangle : L^2 \times L^2 \rightarrow \mathbb{R}$ by

$$\langle f, g \rangle := \int_{-\infty}^{\infty} f(x)g(x)dx.$$

Roughly, the inner product $\langle \cdot, \cdot \rangle$ satisfies the properties in Proposition 7.1.

- 3 When we consider square integrable random variables, we can also introduce the inner product as

$$\langle x, y \rangle := \mathbb{E}[xy] \text{ or } \langle x, y \rangle := \mathbb{E}[(x - \bar{x})(y - \bar{y})] = \text{Cov}(x, y)$$

in the similar manner.

Riesz Representation Theorem (for $\mathbb{R}^n$)

Theorem 7.1 (Riesz Representation Theorem for $\mathbb{R}^n$)

- ① Take $\mathbf{a} \in \mathbb{R}^n$. The function $f_{\mathbf{a}} : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f_{\mathbf{a}}(\mathbf{x}) := \langle \mathbf{a}, \mathbf{x} \rangle \text{ for each } \mathbf{x} \in \mathbb{R}^n$$

is a linear mapping.

- ② (Riesz Representation Theorem) For any given linear mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}$, there exists a unique element $\mathbf{a} \in \mathbb{R}^n$ such that

$$\underbrace{f}_{\text{linear function}}(\mathbf{x}) = \langle \underbrace{\mathbf{a}}_{\text{vector in } \mathbb{R}^n}, \mathbf{x} \rangle \text{ for each } \mathbf{x} \in \mathbb{R}^n.$$

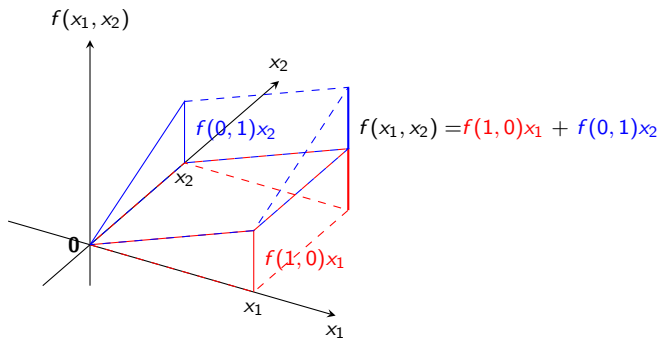
Example.

$$\underbrace{E[U]}_{\text{expected utility}} = \sum_{i=1}^n \underbrace{p_i}_{\text{linear in probability } p(=\mathbf{x})} \underbrace{u_i}_{\text{utility index } u(=\mathbf{a})}$$

Riesz Representation Theorem (for $\mathbb{R}^n$)

Intuition ($n = 2$)

$$f(\underbrace{x_1, x_2}_x) = f(1, 0)x_1 + f(0, 1)x_2 = \left\langle \underbrace{\begin{pmatrix} f(1, 0) \\ f(0, 1) \end{pmatrix}}_{=a}, \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{=x} \right\rangle.$$



Riesz Representation Theorem (for $\mathbb{R}^n$)

Remark 7.3

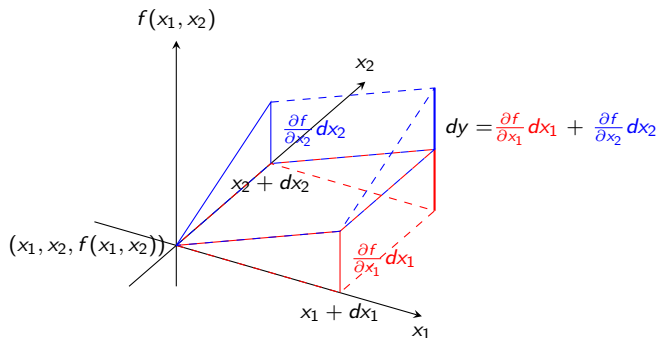
- Applications of Riesz representation theorem include, somewhat roughly, the existence of an expected-utility representation in economics and the existence of pricing kernel (stochastic discount factor) in finance or macroeconomics.
- Also, for a mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with $n = 2$, the (total) differential dy of $y = f(x_1, x_2)$ is represented as an inner product

$$\underbrace{dy}_{\text{linear in } (dx_1, dx_2)} = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 = \left\langle \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix}, \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} \right\rangle = \langle \nabla f, dx \rangle,$$

where $\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix}$ is the gradient of f .

Riesz Representation Theorem (for $\mathbb{R}^n$)

Differentiation: Approximating a function f by a linear function



Matrix Representation Theorem

Matrix Representation Theorem

- ① A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if and only if there exists a unique $m \times n$ matrix A such that for every $x \in \mathbb{R}^n$,

$$T(x) = Ax.$$

- ② Moreover,

$$\underbrace{A}_{m \times n} = \left(\underbrace{T(e^1)}_{m \times 1}, \underbrace{T(e^2)}_{m \times 1}, \dots, \underbrace{T(e^n)}_{m \times 1} \right),$$

where $e^i = (0, \dots, 1, \dots, 0)^T$ is the i -th unit vector.

$$\begin{pmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} & \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \langle a_1, x \rangle \\ \langle a_2, x \rangle \\ \vdots \\ \langle a_m, x \rangle \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}.$$

Matrix Representation Theorem

- We have:

$$\begin{aligned} Ax &= \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} \\ &= x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}. \end{aligned}$$

- So, Ax can also be seen as the linear combination of the column vectors of A with $x^T = (x_1 \ x_2 \ \dots \ x_n)$ being its coefficients.
- The i -th column of A is also expressed as Ae^i .

Matrix Representation Theorem

Proof Sketch (Optional)

If T is linear, then

WTS (i): $T(x) = Ax,$
 $A = (T(e^1), \dots, T(e^n)).$

WTS (ii): A is unique:
If $Ax = T(x) = Bx$ then $A = B.$

$$\begin{aligned} T(x) &= T\left(\sum_{i=1}^n x_i e^i\right) \\ &= \sum_{i=1}^n x_i T(e^i) \\ &= (T(e^1), \dots, T(e^n)) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= Ax. \end{aligned}$$

$\Rightarrow$

$$A = (T(e^1), \dots, T(e^n))$$

$$\begin{aligned} B &= (b^1, \dots, b^n) \\ &= (Be^1, \dots, Be^n) \\ &= (T(e^1), \dots, T(e^n)) \\ &= A \end{aligned}$$

$$\Rightarrow A = B$$

Matrix Representation Theorem

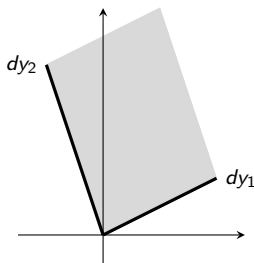
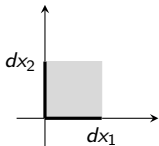
For a mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, say, with $n = m = 2$, we can represent the total differential as

$$\begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix} \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix}, \text{ that is, } dy = Jdx,$$

where

$$J = \begin{pmatrix} (\nabla f_1)^T \\ (\nabla f_2)^T \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix}$$

is referred to as the Jacobian matrix of f .



Also, $\det J$ is the rate at which the area determined by dx_1 and dx_2 is enlarged to the area determined by dy_1 and dy_2 (with $\pm$ sign).

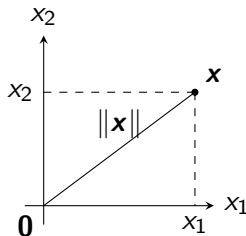
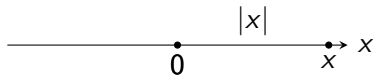
Norm

- When a vector space X is endowed with an inner product, the norm of an element $x \in X$ is defined as $\|x\| := \sqrt{\langle x, x \rangle}$. In $\mathbb{R}^n$:

Definition 7.2

The norm of a vector $x \in \mathbb{R}^n$ is defined as

$$\|x\| := \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i^2}.$$



Cauchy-Schwarz Inequality

Cauchy-Schwarz Inequality: relation between an inner product and the norm

For any $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$,

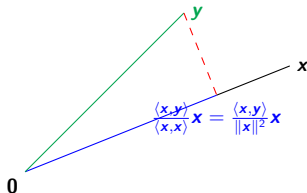
$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

The equality holds if and only if $(\mathbf{x}, \mathbf{y})$ are linearly dependent.

Exercise 7.2

Show that Cauchy-Schwarz inequality holds with equality if and only if either $\mathbf{y} = \mathbf{0}$ or $\mathbf{y} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} \mathbf{x}$.

$$\begin{aligned}\langle \lambda \mathbf{x}, \mathbf{y} - \lambda \mathbf{x} \rangle &= 0 \\ \Rightarrow \lambda \langle \mathbf{x}, \mathbf{y} \rangle &= \lambda^2 \langle \mathbf{x}, \mathbf{x} \rangle \\ \Rightarrow \lambda &= \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle}\end{aligned}$$



$$\begin{aligned}\left\| \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|^2} \mathbf{x} \right\| &\leq \|\mathbf{y}\| \\ \underbrace{\frac{|\langle \mathbf{x}, \mathbf{y} \rangle|}{\|\mathbf{x}\|}}_{\text{}} &\leq \|\mathbf{y}\| \\ \Rightarrow |\langle \mathbf{x}, \mathbf{y} \rangle| &\leq \|\mathbf{x}\| \|\mathbf{y}\|.\end{aligned}$$

Remark/Exercise on Cauchy-Schwarz Inequality

Exercise 7.6 (Optional)

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{y} = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$. Define the mean of $\mathbf{x}$ and $\mathbf{y}$ as

$$\mu_x = \frac{1}{n} \sum_{i=1}^n x_i \text{ and } \mu_y = \frac{1}{n} \sum_{i=1}^n y_i, \text{ respectively.}$$

Define the covariance of $\mathbf{x}$ and $\mathbf{y}$ as

$$\sigma_{x,y} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y).$$

Define the variance of $\mathbf{x}$ and $\mathbf{y}$ as $\sigma_x^2 = \sigma_{x,x}$ and $\sigma_y^2 = \sigma_{y,y}$, respectively. Assume $\sigma_x^2 > 0$ and $\sigma_y^2 > 0$.

Finally, define the (Pearson) correlation coefficient as

$$\rho = \frac{\sigma_{x,y}}{\sigma_x \sigma_y}.$$

Now, prove the following.

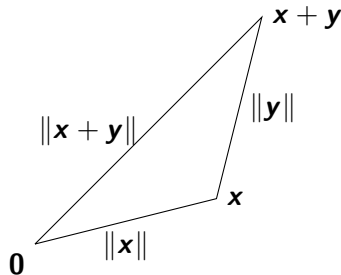
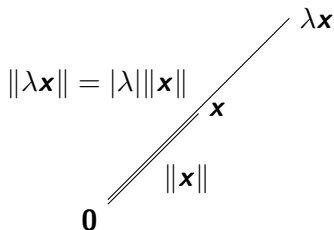
- ① $|\rho| \leq 1$.
- ② $|\rho| = 1$ if and only if $y_i - \mu_y = \frac{\sigma_{x,y}}{\sigma_x^2} (x_i - \mu_x)$ for all $i \in \{1, 2, \dots, n\}$.

Norm

Proposition 7.2

The norm $\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}$ possesses the following properties.

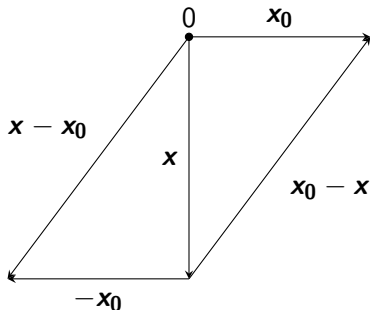
- 1 $\| \mathbf{x} \| \geq 0$.
- 2 $\| \mathbf{x} \| = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
- 3 $\| \lambda \mathbf{x} \| = |\lambda| \| \mathbf{x} \|$.
- 4 $\| \mathbf{x} + \mathbf{y} \| \leq \| \mathbf{x} \| + \| \mathbf{y} \|$.



Norm: An Exercise

Exercise 7.4

Show that, for any $\mathbf{x}_0, \mathbf{x} \in \mathbb{R}^n$, $|||\mathbf{x}|| - ||\mathbf{x}_0||| \leq ||\mathbf{x} - \mathbf{x}_0||$.



Norm

Remark 7.5

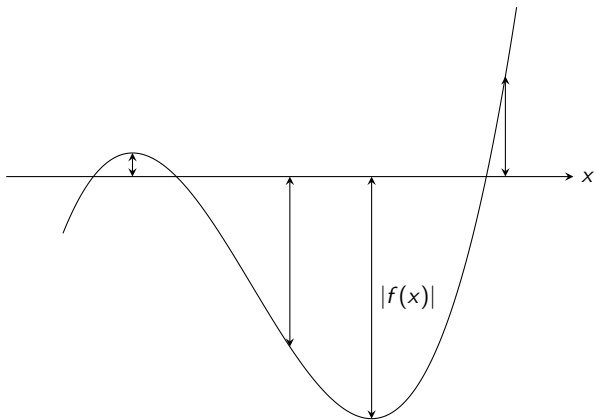
- ❶ A vector space X is a *normed space* if it is endowed with the norm $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfying the properties in Proposition 7.2.
- ❷ An inner product space X (endowed with the inner product $\langle \cdot, \cdot \rangle$) is a normed space when the norm $\|\cdot\| : X \rightarrow \mathbb{R}$ is given by $\|x\| = \sqrt{\langle x, x \rangle}$ for each $x \in X$.
- ❸
 - ▶ As an example of a normed space, consider the set $C[a, b]$ of continuous functions $f : [a, b] \rightarrow \mathbb{R}$ defined on an interval $[a, b]$ in $\mathbb{R}$.
 - ▶ Define the norm $\|\cdot\|$ (or $\|\cdot\|_\infty$) as

$$\|f\| := \sup_{x \in [a, b]} |f(x)|.$$

- ▶ It can be seen that $\|\cdot\|$ is indeed a norm, i.e., satisfies the properties in Proposition 7.2.
- ▶ This norm may be used in dynamic programming.
- ▶ Dynamic programming is one of the very useful techniques to study dynamic economic problems which show up in macroeconomics but also in other fields of economics.

Norm

$$\|f\| := \sup_{x \in [a,b]} |f(x)|.$$



Distance

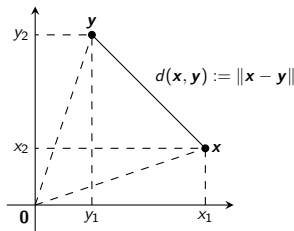
- When a set X is endowed with a norm $\|\cdot\|$, the distance (or the metric) $d : X \times X \rightarrow \mathbb{R}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| \text{ for each } \mathbf{x}, \mathbf{y} \in X.$$

Definition 7.3

In $\mathbb{R}^n$, the distance $d(\mathbf{x}, \mathbf{y})$ between two points $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}.$$

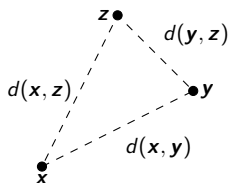
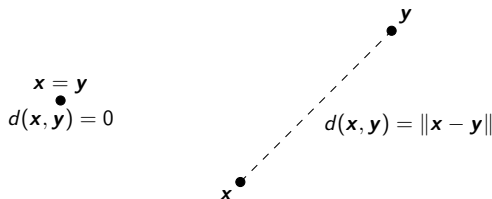


Distance

Proposition 7.3

The distance $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the following properties.

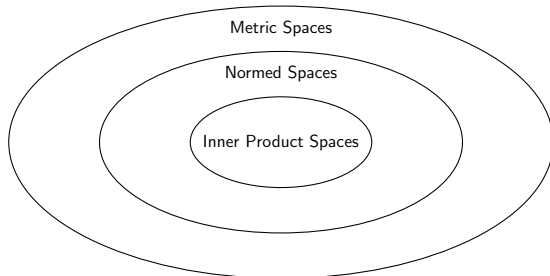
- 1 $d(\mathbf{x}, \mathbf{y}) \geq 0$.
- 2 $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$.
- 3 $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$.
- 4 $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$.



Distance

Remark 7.7

- ① A set X is a *metric space* if it is endowed with a distance $d : X \times X \rightarrow \mathbb{R}$ satisfying the properties in Proposition 7.3.
- ②
 - ▶ A normed space is a metric space, where the metric $d : X \times X \rightarrow \mathbb{R}$ is defined by $d(x, y) = \|x - y\|$.
 - ▶ Since an inner product space is a normed space, it is also a metric space.

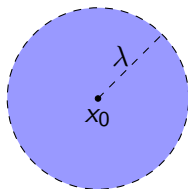


Open Balls

Definition 7.4

Let $x_0 \in \mathbb{R}^n$ and $\lambda > 0$. The *open ball* with center x_0 and radius λ is a set

$$B_\lambda(x_0) := \{x \in \mathbb{R}^n \mid d(x_0, x) < \lambda\}.$$

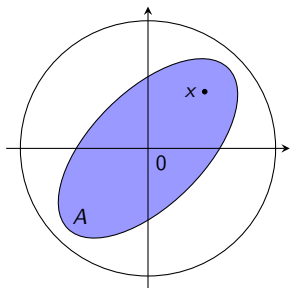


$B_\lambda(x_0)$: the set of $x \in \mathbb{R}^n$ such that
the distance between x_0 and x is less than λ .

Bounded Sets

Definition 7.8

A set A (in $\mathbb{R}^n$) is *bounded* if there exists a number $\alpha > 0$ such that $x \in A$ implies $\|x\| \leq \alpha$.



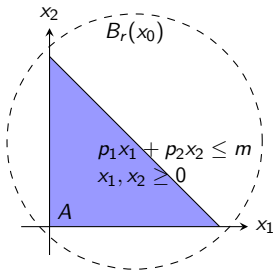
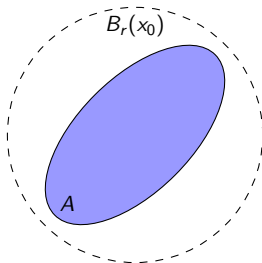
There exists some “bound” $\alpha > 0$ such that the norm $\|x\|$ of x (or the distance between x and 0) is bounded by α : $\|x\| \leq \alpha$.

Bounded Sets: An Exercise

Exercise 7.20

Show that the following are equivalent.

- 1 A subset A of $\mathbb{R}^n$ is bounded.
- 2 There exist $x_0 \in \mathbb{R}^n$ and $r > 0$ such that $A \subseteq B_r(x_0)$.
(A is included in an open ball.)

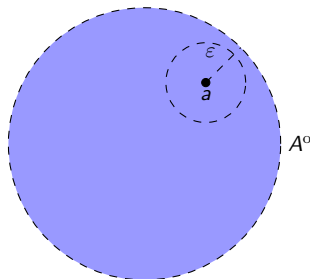
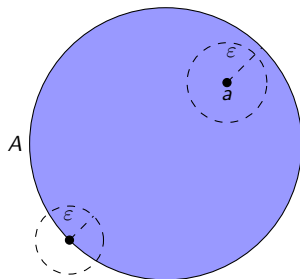


Open and Closed Sets

Definition 7.5

Let A be a subset of $\mathbb{R}^n$.

- ① A point $a \in \mathbb{R}^n$ is *interior* to the set A if there exists $\varepsilon > 0$ such that $B_\varepsilon(a) \subseteq A$.
- ② The interior of the set A , denoted by A° , is the set of interior points of A .
- ③ By definition, $A^\circ \subseteq A$.
- ④ We call A an *open set* if $A = A^\circ$. That is, for any $a \in A$, there exists $\varepsilon > 0$ such that $B_\varepsilon(a) \subseteq A$.

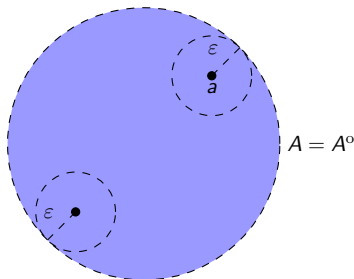
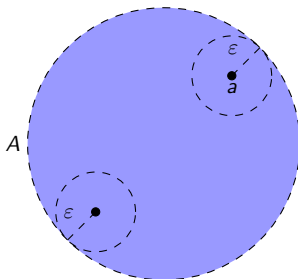


Open and Closed Sets

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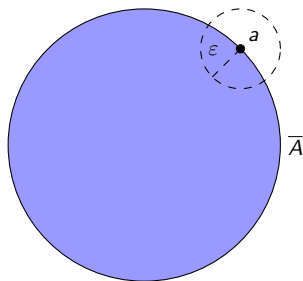
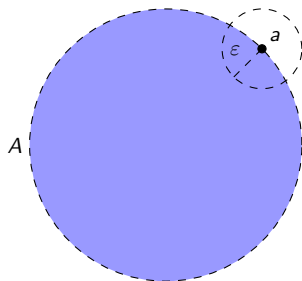


Open and Closed Sets

Definition 7.5 (Cont'd)

Let A be a subset of $\mathbb{R}^n$.

- ②
 - ① The point $a \in \mathbb{R}^n$ is a *point of closure* of the set A if, for any given $\varepsilon > 0$, $A \cap B_\varepsilon(a) \neq \emptyset$.
 - ② The closure of the set A , denoted by $\overline{A}$, is the set of points of closure of A .
 - ③ By definition, $A \subseteq \overline{A}$. If the set A satisfies $A = \overline{A}$, we call A a *closed set*.

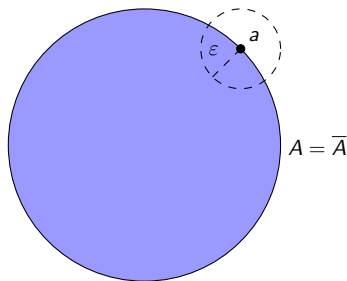
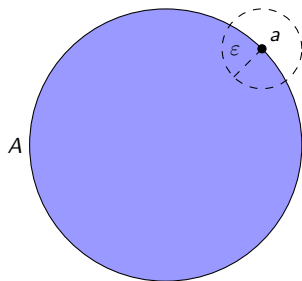


Open and Closed Sets

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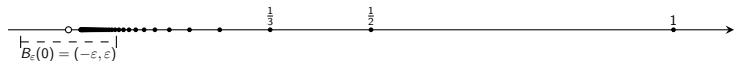
Open and Closed Sets

Example 7.1

In $\mathbb{R}$, the distance function d is given by $d(x, y) = |x - y|$ for each $x, y \in \mathbb{R}$. Now, consider the set

$$A = \left\{ 1, \frac{1}{2}, \dots, \frac{1}{n}, \dots \right\} = \left\{ x \in \mathbb{R} \mid x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \right\}.$$

We have $0 \in \bar{A}$: for any $\varepsilon > 0$, $\underbrace{\frac{1}{n} \in A \cap B_\varepsilon(0)}_{A \cap B_\varepsilon(0) \neq \emptyset}$ for some $n \in \mathbb{N}$.



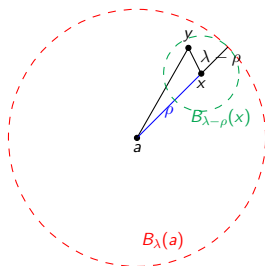
Open and Closed Sets

Proposition 7.4

In $\mathbb{R}^n$, every open ball $B_\lambda(a)$ is an open set.

Proof.

Take any point $x \in B_\lambda(a)$. We show that x belongs to the interior of $B_\lambda(a)$. Take $\rho = d(a, x) < \lambda$. For any $y \in B_{\lambda-\rho}(x)$, we have $d(x, y) < \lambda - \rho$. Thus, $d(a, y) \leq d(a, x) + d(x, y) < \rho + \lambda - \rho = \lambda$. Thus, $y \in B_\lambda(a)$. Hence, $B_{\lambda-\rho}(x) \subseteq B_\lambda(a)$, that is, x is in the interior of $B_\lambda(a)$. □



Open and Closed Sets: Exercises

Proposition 7.5

A point $a \in \mathbb{R}^n$ is a point of closure of a set A if and only if a is not in the interior of A^c . In other words,

$$(\overline{A})^c = (A^c)^\circ.$$

Corollary 7.1

The complement of a closed set is an open set, and vice versa.

$$\begin{array}{lll} A: \text{ open} & \iff & A^c: \text{ closed} \\ B: \text{ closed} & \iff & B^c: \text{ open} \end{array}$$

Proposition 7.6

Let A be a set in $\mathbb{R}^n$.

- 1 The interior A° itself is open.
- 2 The closure $\overline{A}$ itself is closed.

Open and Closed Sets: Exercise

Proposition 7.7

The collection of open sets in $\mathbb{R}^n$ has the following properties.

- 1 Every union of open sets is open.
- 2 Every finite intersection of open sets is open.
- 3 The sets $\emptyset$ and $\mathbb{R}^n$ are open.

Remark 7.12

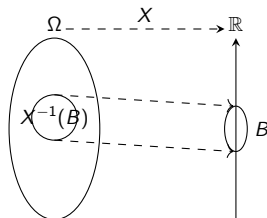
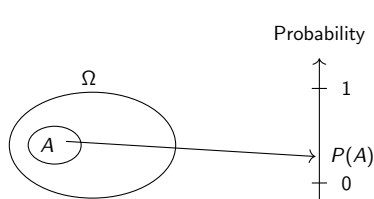
Let X be a set. Suppose that a collection $\mathcal{O}$ of subsets of X satisfies the three properties in Proposition 7.7:

- 1 The union of any sub-collection of $\mathcal{O}$ is in $\mathcal{O}$.
- 2 The intersection of any finite sub-collection of $\mathcal{O}$ is in $\mathcal{O}$.
- 3 The collection $\mathcal{O}$ contains the sets $\emptyset$ and X .

Then, we call the collection $\mathcal{O}$ a *topology* on X , and a subset A of X is *open* if $A \in \mathcal{O}$.

Open and Closed Sets: Remark

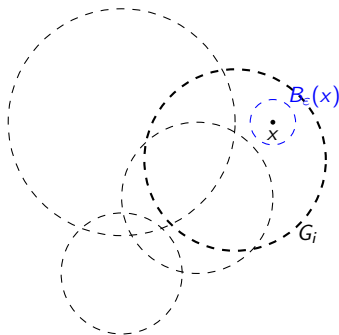
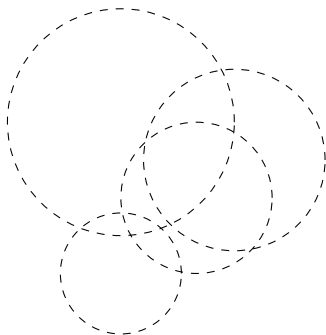
- 1 Previously, we assigned, to every subset A of an at-most countable set Ω , the probability $P(A)$.
 - When Ω is a continuum (e.g., $\Omega = \mathbb{R}$ or $\Omega = [0, 1]$), we need to specify a collection $\mathcal{F}$ of subsets of Ω (a “ σ -algebra”) for which a probability measure $P : \mathcal{F} \rightarrow [0, 1]$ is defined.
 - Each $A \in \mathcal{F}$ is referred to as a “measurable set.”
 - On $\mathbb{R}$ (or its subset), the natural σ -algebra (“Borel σ -algebra”) is generated from the collection of open sets $\mathcal{O}$.
- 2
 - $f : X \rightarrow Y$ is continuous if, for any open set B in Y , $f^{-1}(B)$ is an open set in X .
 - $X : \Omega \rightarrow \mathbb{R}$ is a random variable if, for any “measurable set” B in $\mathbb{R}$, $X^{-1}(B)$ is a measurable set in Ω .



Open and Closed Sets: Solution

Proof of Proposition 7.7 (1)

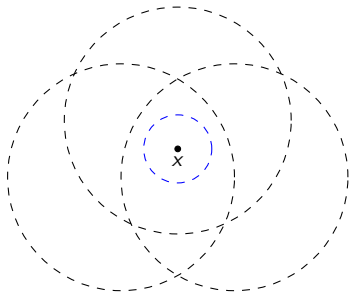
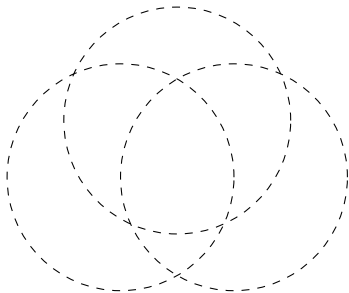
Let $(G_i)_{i \in I}$ be a collection of open sets. If $x \in G = \bigcup_{i \in I} G_i$, then there is $i \in I$ such that $x \in G_i$. Since G_i is open, there is $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq G_i \subseteq G$. Thus, G is open.



Open and Closed Sets: Solution

Proof of Proposition 7.7 (2)

Let $(G_i)_{i=1}^n$ be a finite collection of open sets. If $x \in G = \bigcap_{i=1}^n G_i$, then $x \in G_i$ for all $i \in \{1, \dots, n\}$. For each i , there exists $\varepsilon_i > 0$ such that $B_{\varepsilon_i}(x) \subseteq G_i$. Now, take $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$. Then, $B_\varepsilon(x) \subseteq B_{\varepsilon_i} \subseteq G_i$ for each $i \in \{1, \dots, n\}$, and thus $B_\varepsilon(x) \subseteq \bigcap_{i=1}^n G_i = G$. Thus, G is open.



Open and Closed Sets: Solution

Proof of Proposition 7.7 (3)

For $\mathbb{R}^n$, if $x \in \mathbb{R}^n$ then $B_\varepsilon(x) \subseteq \mathbb{R}^n$ for some $\varepsilon > 0$ (for example, take $\varepsilon = 1$).

For the empty set $\emptyset$, if $x \in \emptyset$ then vacuously $B_\varepsilon(x) \subseteq \emptyset$ for some $\varepsilon > 0$. □

Remark: In mathematics, the statement of the form " $P \Rightarrow Q$ " (P implies Q) is equivalent to $(\neg P) \vee Q$ (either $\neg P$ (not P) or Q).

P	Q	$(\neg P)$	$(\neg P) \vee Q$	$P \Rightarrow Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

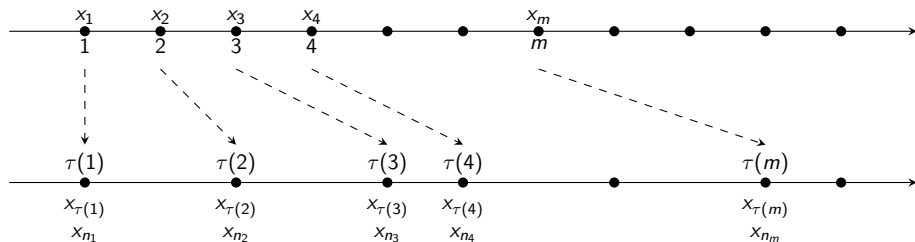
Sequences

Definition 7.10

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$, and let $\tau : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function (i.e., $m < n$ implies $\tau(m) < \tau(n)$).

Then, we call $(x_{\tau(n)})_{n \in \mathbb{N}}$ a subsequence of $(x_n)_{n \in \mathbb{N}}$.

We often denote by $(x_{n_k})_{k \in \mathbb{N}}$ a subsequence of $(x_n)_{n \in \mathbb{N}}$.

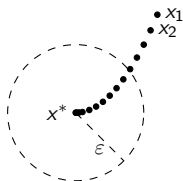


Sequences

Definition 7.11

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$.

- The sequence tends to a limit x^* , denoted by $x^* = \lim_{n \rightarrow \infty} x_n$ or $x_n \rightarrow x^*$ (as $n \rightarrow \infty$), if for any given number $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $n \geq n_0$ implies $d(x^*, x_n) < \varepsilon$.
- The sequence is *convergent* if there exists $x^* \in \mathbb{R}^n$ such that $x^* = \lim_{n \rightarrow \infty} x_n$.



Sequences: Exercises

Exercise 7.30

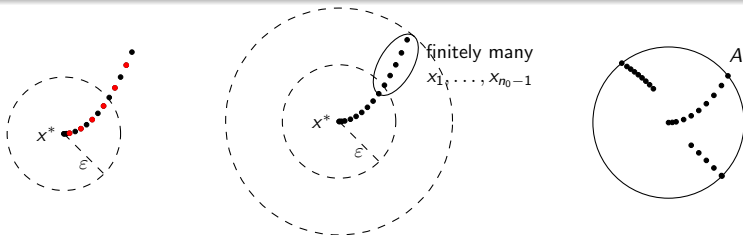
Show that if a sequence $(x_n)_{n \in \mathbb{N}}$ tends to a limit x then any subsequence tends to x .

Exercise 7.31

Show that if $(x_n)_{n \in \mathbb{N}}$ is a convergent sequence in $\mathbb{R}^n$, then it is bounded, i.e., there exists $M > 0$ such that $\|x_n\| \leq M$ for all $n \in \mathbb{N}$.

Proposition 7.11

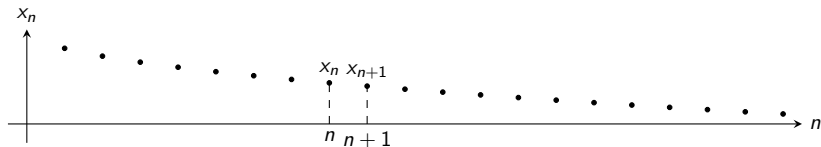
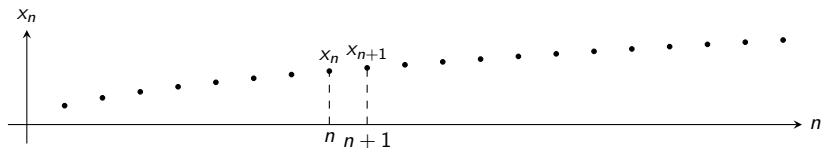
A set A is closed in $\mathbb{R}^n$ if and only if, for any convergent sequence $(x_n)_{n \in \mathbb{N}}$ in A , the limit lies in A .



Bounded Sequences

Definition 7.13

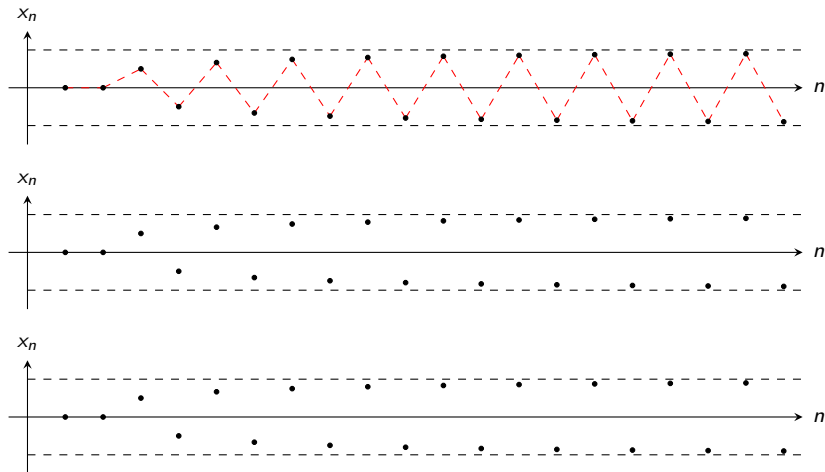
- ① A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is *increasing* if $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$.
It is *strictly increasing* if $x_{n+1} > x_n$ for all $n \in \mathbb{N}$.
- ② A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is *decreasing* if $x_{n+1} \leq x_n$ for all $n \in \mathbb{N}$.
It is *strictly decreasing* if $x_{n+1} < x_n$ for all $n \in \mathbb{N}$.



Bounded Sequences

Proposition 7.12

Every sequence in $\mathbb{R}$ has either an increasing subsequence or a decreasing subsequence, and possibly both.



Bounded Sequences

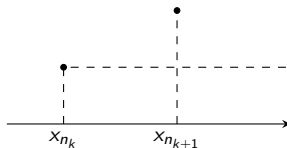
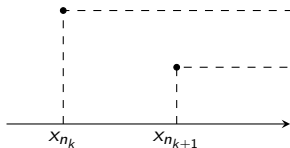
Proof of Proposition 7.12 (Optional).

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. Define

$$S := \{m \in \mathbb{N} \mid x_m > x_n \text{ for all } n > m\}.$$

If S is an infinite set, there is a decreasing subsequence $(x_{n_k})_{k \in \mathbb{N}}$. If S is a finite set, then there is an increasing subsequence $(x_{n_k})_{k \in \mathbb{N}}$. □

- The left panel illustrates the case in which S is an infinite set. It depicts two elements x_{n_k} and $x_{n_{k+1}}$ in S . As $x_{n_k} \in S$, any element x_n with $n > n_k$ satisfies $x_n > x_{n_{k+1}}$. Since there are infinitely many elements $(x_{n_k})_{k \in \mathbb{N}}$ in S , it follows that one can take a decreasing subsequence $(x_{n_k})_{k \in \mathbb{N}}$.
- The right panel illustrates the case in which S is a finite set. This means that, for any x_{n_k} , there exists $x_{n_{k+1}}$ with $x_{n_{k+1}} > x_{n_k}$. Thus, one can take an increasing subsequence $(x_{n_k})_{k \in \mathbb{N}}$.



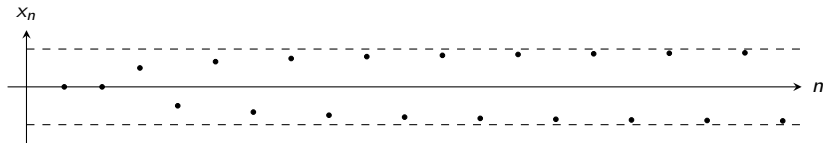
Bounded Sequences

Proposition 7.13

Every bounded sequence in $\mathbb{R}$ has at least one convergent subsequence.

Proof.

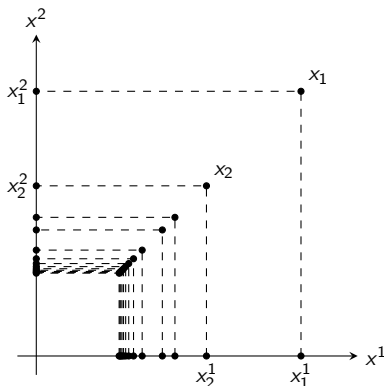
Every bounded sequence $(x_n)_{n \in \mathbb{N}}$ has either an increasing (bounded) subsequence $(x_{n_k})_{k \in \mathbb{N}}$ converging to $\sup_{k \in \mathbb{N}} x_{n_k}$ or a decreasing (bounded) subsequence converging to $\inf_{k \in \mathbb{N}} x_{n_k}$. □



Bounded Sequences

Proposition 7.14

A sequence $(x_m)_{m \in \mathbb{N}}$ in $\mathbb{R}^n$ is convergent if and only if $(x_m^i)_{m \in \mathbb{N}}$ in $\mathbb{R}$ is convergent for each $i \in \{1, 2, \dots, n\}$, where $x_m = (x_m^1, x_m^2, \dots, x_m^n)$.



Continuous Functions

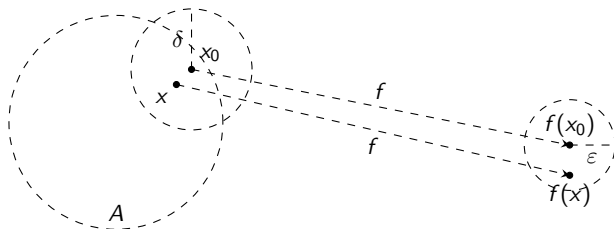
Definition 7.15

Let A be a set in $\mathbb{R}^n$, and let $f : A \rightarrow \mathbb{R}^m$ be a mapping.

- 1 The mapping f is *continuous (with respect to A) at a point $x_0 \in A$* if, for any given number $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in A \text{ and } d(x_0, x) < \delta \text{ imply } d(f(x_0), f(x)) < \varepsilon.$$

- 2 The function is *continuous in A (with respect to A)* if it is continuous at every point in A .



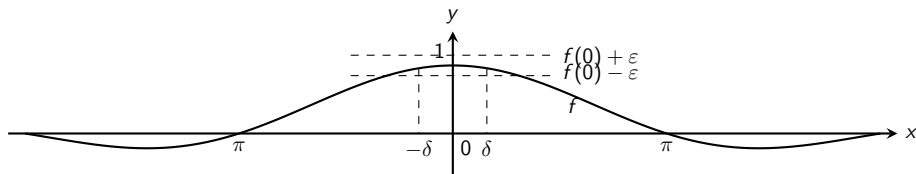
Limits

Example 7.9

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

is continuous in $\mathbb{R}$ (especially at $x = 0$).



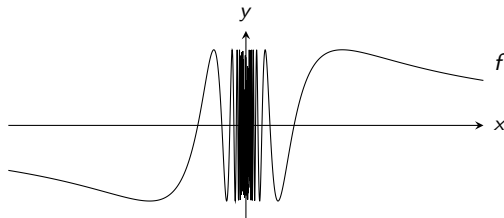
Limits

Example 7.11

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is not continuous at $x = 0$.



- The function f oscillates between -1 and 1 infinitely many times as x gets closer to 0 .

Continuous Functions

Example 7.12

Every constant function is continuous.

————— f

Proposition 7.17

Let $a \in \mathbb{R}^n$. The linear function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(x) = \langle a, x \rangle \text{ for each } x \in \mathbb{R}^n$$

is continuous.

Proof (only when time permits).

If $a = \mathbf{0}$, then f is a constant function, and thus it is continuous. Assume $a \neq \mathbf{0}$. Take $x_0 \in \mathbb{R}^n$. Take $\varepsilon > 0$. Then, let $\delta = \frac{\varepsilon}{\|a\|} > 0$. If $\|x - x_0\| \leq \delta$ then it follows from Cauchy-Schwarz inequality (Lemma 7.1) that

$$|f(x) - f(x_0)| = |\langle a, x \rangle - \langle a, x_0 \rangle| = |\langle a, x - x_0 \rangle| \leq \|a\| \|x - x_0\| < \|a\| \delta = \varepsilon.$$

Thus, f is continuous at every x_0 , that is, f is continuous. □

Continuous Functions: Some Facts

Corollary 7.3

Any linear mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous.

Proposition 7.18

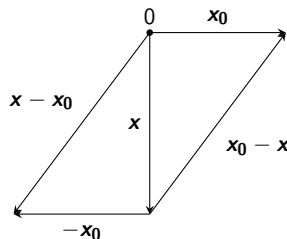
In $\mathbb{R}^n$, the norm $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous.

Hint:

Exercise 7.4

Show that, for any $\mathbf{x}_0, \mathbf{x} \in \mathbb{R}^n$,

$$||\mathbf{x}|| - ||\mathbf{x}_0|| \leq ||\mathbf{x} - \mathbf{x}_0||.$$



Continuous Functions

Example 7.16 (1)

- Let $f : \mathbb{R}^p \times \mathbb{R}^q \rightarrow \mathbb{R}^m$ be a continuous mapping.
- For any $y_0 \in \mathbb{R}^q$, the mapping $f_{y_0} : \mathbb{R}^p \rightarrow \mathbb{R}^m$ defined by

$$f_{y_0}(x) = f(x, y_0) \text{ for all } x \in \mathbb{R}^p$$

is continuous.

- Similarly, for any $x_0 \in \mathbb{R}^p$, the mapping $f_{x_0} : \mathbb{R}^q \rightarrow \mathbb{R}^m$ defined by

$$f_{x_0}(y) = f(x_0, y) \text{ for all } y \in \mathbb{R}^q$$

is continuous.

- In other words, every function continuous with respect to a pair of two variables is separately continuous with respect to each of the two variables.

Continuous Functions

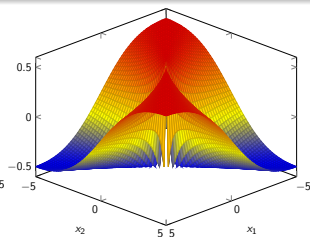
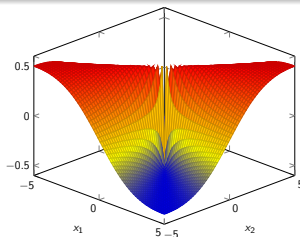
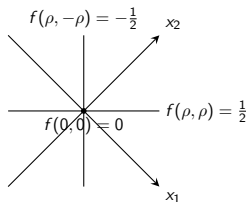
Example 7.16 (2)

- However, the converse is not always true. Consider $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}.$$

The function is separately continuous with respect to the two variables x_1 and x_2 , but it is not continuous at the point $(0, 0)$. For example, for ρ with $|\rho| > 0$ small, we have

$$f(\rho, \rho) = \frac{1}{2} \text{ and } f(\rho, -\rho) = -\frac{1}{2}.$$

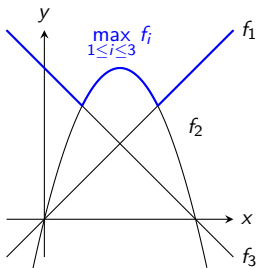


Continuous Functions: Some facts

A composite of continuous functions is continuous.

Exercise 7.34

Let A be a subset of $\mathbb{R}^n$. Let $f_i : A \rightarrow \mathbb{R}$ be a continuous function for each $i \in \{1, 2, \dots, m\}$. Show that $f := \max_{1 \leq i \leq m} f_i$ is continuous.

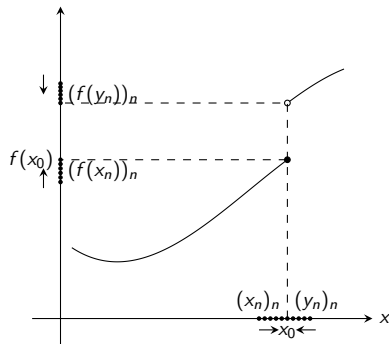
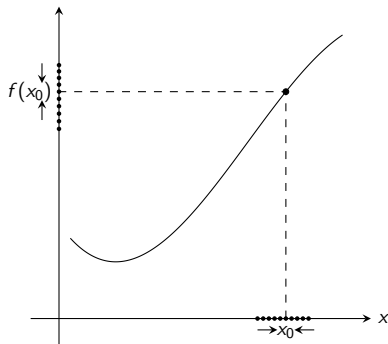


Continuous Functions

Proposition 7.22

Let A be a subset of $\mathbb{R}^n$, and let $f : A \rightarrow \mathbb{R}^m$ be a mapping. The function f is continuous at $x_0 \in A$ if and only if, for *any* sequence $(x_n)_{n \in \mathbb{N}}$ converging to x_0 , the sequence $(f(x_n))_{n \in \mathbb{N}}$ converges to $f(x_0)$, that is,

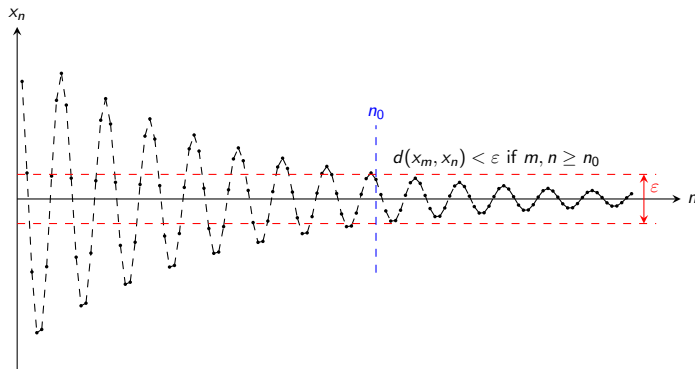
$$f\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} f(x_n).$$



Completeness of a Metric Space

Definition 8.8

Let X be a metric space. A sequence $(x_n)_{n \in \mathbb{N}}$ in X is a Cauchy sequence if, for any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that if $m, n \geq n_0$ then $d(x_m, x_n) < \varepsilon$.



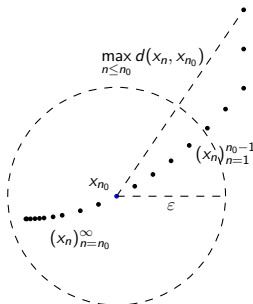
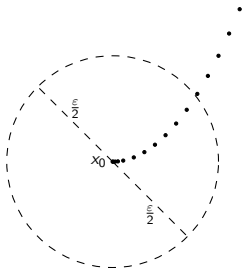
Completeness of a Metric Space: Exercises

Proposition 8.14

Let X be a metric space. Any convergent sequence $(x_n)_{n \in \mathbb{N}}$ in X is a Cauchy sequence.

Proposition 8.15

Let X be a metric space. Any Cauchy sequence $(x_n)_{n \in \mathbb{N}}$ in X is bounded.



Completeness of a Metric Space

Definition 8.9

A metric space X is *complete* if any Cauchy sequence $(x_n)_{n \in \mathbb{N}}$ has the limit $x = \lim_{n \rightarrow \infty} x_n$ in X .

Generally: Convergent sequence $\implies$ Cauchy sequence
 $\not\Leftarrow$

X Complete: Convergent sequence $\iff$ Cauchy sequence

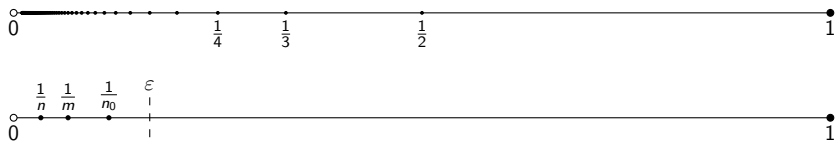
When X is complete:

To show that $(x_n)_{n \in \mathbb{N}}$ has a limit, it suffices to show that $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence.

Completeness of a Metric Space

Example

The space $X = (0, 1]$ is not complete. While $(x_n)_{n \in \mathbb{N}}$ defined by $x_n = \frac{1}{n}$ is a Cauchy sequence, its limit $0 = \lim_{n \rightarrow \infty} x_n$ does not belong to X .



For any $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $d(\frac{1}{n}, \frac{1}{m}) < \varepsilon$ if $m, n \geq n_0$.

Completeness of a Metric Space: Exercises

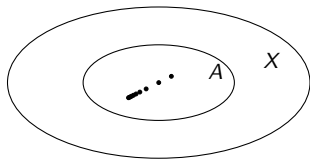
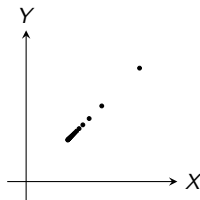
Proposition 8.18

Let X and Y be complete metric spaces. Then, $X \times Y$ is a complete metric space.

- $\mathbb{R}$, $\mathbb{R}^2$, and $\mathbb{R}^n$ are complete.

Proposition 8.19

Let X be a complete metric space. A subset A is complete if and only if A is closed.

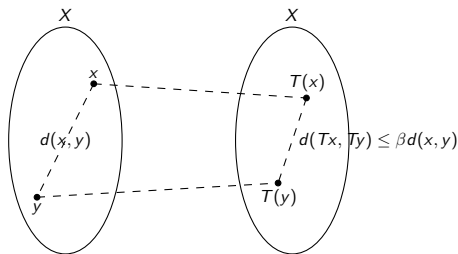


Contraction Mapping Theorem

Definition 8.10

A mapping T from a metric space (X, d) into itself is called a *contraction* (mapping with modulus β) if there exists $\beta \in [0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in X.$$



Contraction Mapping Theorem

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$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in X.$$

Theorem 8.3 (Banach Contraction Mapping Theorem)

Let (X, d) be a complete metric space. Let $T : X \rightarrow X$ be a contraction. Then, T has a unique fixed point (i.e., a unique $x^* \in X$ such that $T(x^*) = x^*$).

- 1 The unique fixed point can be obtained as $x = \lim_{n \rightarrow \infty} x_n$, where $x_n = Tx_{n-1}$ with $x_0 \in X$ being an arbitrary point in X .
- 2 The distance between the unique fixed point x and the n -th iteration is

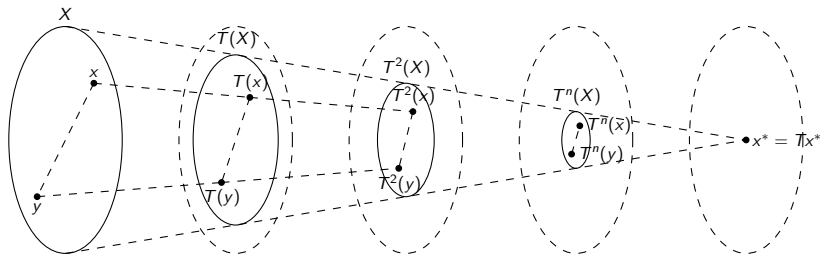
$$d(x_n, x) \leq \frac{\beta^n}{1 - \beta} d(x_0, Tx_0).$$

Contraction Mapping Theorem

Remark 8.5

- Banach contraction mapping theorem is a powerful tool.
- For example, it is usually used in proving
 - ▶ the existence and uniqueness of solutions to certain ordinary differential equations,
 - ▶ the inverse function theorem,
 - ▶ the implicit function theorem,
 - ▶ the existence and uniqueness of solutions to value function iteration, policy iteration, and policy evaluation of certain reinforcement learning algorithms, and,
 - ▶ last but not least, the existence and uniqueness of solutions to dynamic macroeconomic models (see Ljungqvist and Sargent, 2018; and Stokey, Lucas, and Prescott, 1989).

Contraction Mapping Theorem



- 1 Starting with $x_0 \in X$, define $x_n := T^n x_0$.
- 2
 - ▶ $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence.
 - ▶ $\Rightarrow x^* := \lim_{n \rightarrow \infty} x_n$ (X is complete).
- 3 x^* is a fixed point: $Tx^* = T(\lim_n x_n) = \lim_n T(x_n) = x^*$.
- 4 Uniqueness: if $x^* = Tx^*$ and $y^* = Ty^*$ then

$$d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta d(x^*, y^*) \Rightarrow x^* = y^*.$$

Contraction Mapping Theorem

Proof of Theorem 8.3: Step 1

Let $x_0 \in X$ be any point.

We recursively define

$$x_n := Tx_{n-1} \text{ for each } n \in \mathbb{N}.$$

We show that $x := \lim_{n \rightarrow \infty} x_n$ is well-defined and indeed a fixed point.

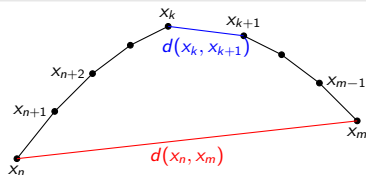
Contraction Mapping Theorem

Proof of Theorem 8.3: Step 2

For any $n, m \in \mathbb{N}$ with $m > n$, we have

$$\begin{aligned} d(x_n, x_m) &\leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) \\ &\leq \sum_{k=n}^{m-1} \beta^k d(x_0, x_1) = \beta^n \frac{1 - \beta^{m-n}}{1 - \beta} d(x_0, x_1) \\ &\leq \frac{\beta^n}{1 - \beta} d(x_0, x_1), \end{aligned}$$

where the first inequality follows from the repeated use of the triangle inequality and the second inequality follows from the repeated use of the definition of a contraction mapping.



$$\begin{aligned} d(x_k, x_{k+1}) &= d(Tx_{k-1}, Tx_k) \\ &\leq \beta d(x_{k-1}, x_k) \\ &\leq \dots \\ &\leq \beta^k d(x_0, x_1). \end{aligned}$$

Contraction Mapping Theorem

Proof of Theorem 8.3: Step 2 (Cont'd)

For any $\varepsilon > 0$, there is $n_0 \in \mathbb{N}$ such that

$$\frac{\beta^n}{1 - \beta} d(x_0, x_1) < \varepsilon \text{ for any } n \geq n_0.$$

Then, we have

$$d(x_n, x_m) < \varepsilon \text{ for any } m, n \geq n_0.$$

This implies that the sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence. Since X is complete, the sequence $(x_n)_{n \in \mathbb{N}}$ is a convergent sequence in X . Thus, $x^* := \lim_{n \rightarrow \infty} x_n \in X$ exists.

Contraction Mapping Theorem

Proof of Theorem 8.3: Step 3

Next, we show that x^* is a fixed point of T .

In order to do so, we first show that T is continuous.

For any $\varepsilon > 0$, let $\delta := \varepsilon/\beta (> 0)$. If $d(x, y) < \delta$, then

$$d(Tx, Ty) \leq \beta \underbrace{d(x, y)}_{< \delta} < \varepsilon.$$

Hence, T is continuous.

Contraction Mapping Theorem

Proof of Theorem 8.3: Step 3 (Cont'd)

Now, since T is continuous, we have

$$Tx^* = T \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

Hence, we obtain $x^* = Tx^*$, that is, x^* is a fixed point.

Contraction Mapping Theorem

Proof of Theorem 8.3: Step 4

Finally, we show that T has a **unique** fixed point.

If x^* and y^* are fixed points of T , then we have

$$d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta d(x^*, y^*),$$

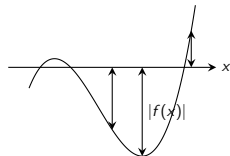
which implies that $x^* = y^*$ (otherwise, we would get $1 \leq \beta < 1$, a contradiction).

Thus, T has the unique fixed point $x^* \in X$.

Blackwell's Condition

- Let $B(\mathbb{R}^n, \mathbb{R})$ be the set of bounded functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$.
- Define the function $\| \cdot \| : B(\mathbb{R}^n, \mathbb{R}) \rightarrow \mathbb{R}$ by

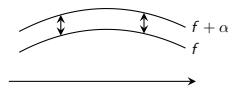
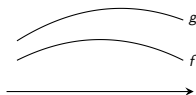
$$\|f\| := \sup_{x \in \mathbb{R}^n} |f(x)|.$$



It can be shown that $\| \cdot \|$ is a norm on $B(\mathbb{R}^n, \mathbb{R})$. This norm is called the sup norm.

- For any $f, g \in B(\mathbb{R}^n, \mathbb{R})$, we denote by $f \leq g$ or $f(\cdot) \leq g(\cdot)$ if $f(x) \leq g(x)$ for all $x \in \mathbb{R}^n$.
- For any $f \in B(\mathbb{R}^n, \mathbb{R})$ and $\alpha \in \mathbb{R}$, define $f + \alpha \in B(\mathbb{R}^n, \mathbb{R})$ by

$$(f + \alpha)(x) := f(x) + \alpha \text{ for all } x \in \mathbb{R}^n.$$



Blackwell's Condition

Proposition 8.20 (Blackwell's Condition)

A mapping $T : \overbrace{B(\mathbb{R}^n, \mathbb{R})}^{=X} \rightarrow B(\mathbb{R}^n, \mathbb{R})$ is a contraction with modulus $\beta \in [0, 1)$ if it satisfies the following two conditions:

- ① Monotonicity: for any $f, g \in B(\mathbb{R}^n, \mathbb{R})$,

$$f \leq g \text{ implies } Tf \leq Tg.$$

- ② Discounting: for any $f \in B(\mathbb{R}^n, \mathbb{R})$ and for any $\alpha \geq 0$,

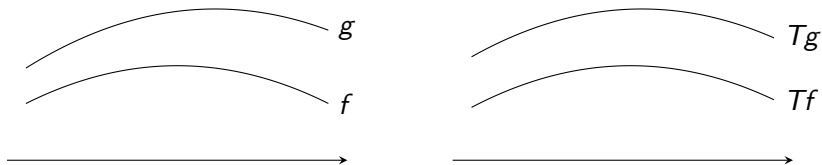
$$T(f + \alpha)(x) \leq Tf(x) + \beta\alpha \text{ for all } x \in \mathbb{R}^n.$$

$$\begin{array}{ccccc} & & Tf & \longrightarrow & T(f) + \beta\alpha \\ & \nearrow & & & \text{VI} \\ f & \longrightarrow & f + \alpha & \longrightarrow & T(f + \alpha) \end{array}$$

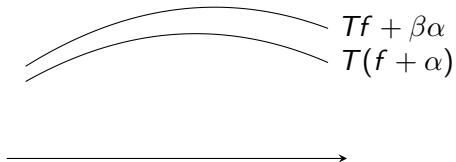
Proof: Exercise 8.24.

Blackwell's Condition

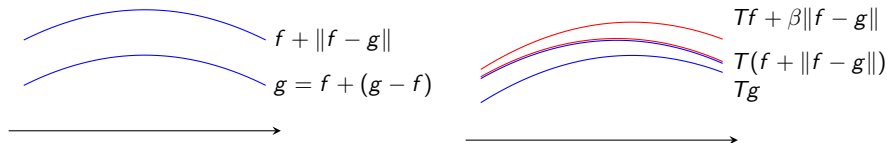
Monotonicity



Discounting



Blackwell's Condition: Solution



$$\textcircled{1} \quad g = f + (g - f) \leq f + \underbrace{\|f - g\|}_{=\sup_x |f(x) - g(x)|}.$$

$\textcircled{2}$

$$\begin{aligned} Tg &\leq T(f + \|f - g\|) && \text{(Monotonicity)} \\ &\leq Tf + \beta\|f - g\|. && \text{(Discounting)} \end{aligned}$$

$$\textcircled{3} \Rightarrow Tg \leq Tf + \beta\|f - g\|.$$

$$\textcircled{4} \quad f \longleftrightarrow g: Tf \leq Tg + \beta\|f - g\|.$$

$$\textcircled{5} \Rightarrow |Tf(x) - Tg(x)| \leq \beta\|f - g\| \text{ for all } x.$$

$$\textcircled{6} \Rightarrow \|Tf - Tg\| \leq \beta\|f - g\|.$$

Additional Definitions

- We have mainly restricted attention to finite dimensional spaces (i.e., $\mathbb{R}^n$).
- The completeness of $\mathbb{R}$ is (equivalent to) an axiom of the real number system.
- The notion of completeness plays a more important role in an infinite dimensional space (e.g., the set of bounded functions $B(\mathbb{R}^n, \mathbb{R})$ in dynamic programming).

Definition 7.12

Let X be a set endowed with a norm $\|\cdot\|$.

The normed (vector) space X is a *Banach space* if it is complete.

Definition 7.13

Let X be a set endowed with an inner product $\langle \cdot, \cdot \rangle$.

The inner product space X is a *Hilbert space* if it is complete.